Test report 24: solidcore update f4f05b1 → 55d0e76

2026-11-12 · mcad commit `adaa3f9` on `ccr-abb3f607-jpendu` · solidcore `tangency-labs/solidcore` main @ `55d0e76` (2026-11-11, crate 1.0.1) · 4 cores, Linux · Rust 1.97 (wasm check 1.98)

Build	Rust tests	diff-reference	Regressions	Eval time, 18 models
ok	**757 / 757**	**2590 / 2604**	**1**	**×1.065**
Rust 1.97 still enough; 0 warnings	after fixes (workspace 647, cad-cli 110)	same as SMLib (before 2596)	wall_hook fillet (upstream #2863)	corpus ×1.16

1 Result

- **Update done.** The pin moved by **527 upstream commits**. Upstream rewrote main's history: `f4f05b1` is no ancestor of `55d0e76`; its tree is `823953ed6` in the new history (`git log f4f05b1..55d0e76` shows 2973 commits, which is wrong).
- **One build break:** the crates `step-brep` and `step-import` are deleted upstream. Polyhedra are now built with solidcore's `BrepConstructor`; all polyhedron nodes keep their verdicts.
- **Three behaviour changes needed adapter work** (blend corners, shell, typed errors), see section 2.
- **One regression remains:** `tests/models/everyday/wall_hook.mcad`, `union(fillet = 2.5)` of a swept tube and a sphere fails (upstream #2863). No workaround on our side; listed in `REFERENCE_EXCEPTIONS`.
- **One gain:** `f12_shell_loft_square_circle` now shells (SMLib failed).
- **Slightly slower:** +6.5 % over 18 models, +16 % over the corpus; trivet ×1.71.

2 Upstream changes and our changes

Upstream	Change	Effect on mcad	Our change
#14038, #14194	`step-brep`, `step-import` deleted; STEP reading in `solidcore-step` (no body builder)	build break in `make_polyhedron`	`polyhedron.rs`: planar faces topology-first via `BrepConstructor` (plane per face, shared line edges and vertices, UV trims)
#13790 and parts	failures carry typed reasons instead of bare `Upstream` statuses	error wording	`lib.rs`: status read via `upstream_status`; manual 17 updated
#2708	blend corner always on: corner of three equal rounds is a `BallPatch`	186 corner faces became approximated B-splines: no radius in `measure`, B-splines in STEP	`brep.rs`: `BallPatch` → neutral sphere
#2769	offset surfaces extended by the offset's reach	`can_shell` (box, `$fillet = 4`, wall 1.5) fails merging the offsets	`offset.rs`: one retry with `merge_results: false`, kept only if it passes the existing shell checks
#2863	fillet solver's start guess from the edge use's UV trim	**wall_hook fillet 2.5 fails (2.25 builds)**	none; exception in the reference test
#14039	certified offset, fold refusal by default	thicken of a peak refused naming the wall	test accepts the named refusal
#14875, #14808, #14984	mass integration	1431 nodes change volume, ≤ 5.6e-6 relative	colour volume test to 1e-9 instead of bit-exact

Upstream	Change	Effect on mcad	Our change
#14799, #15020, retired flags	body minus own copy empty; warnings with results; flags removed	none observed	–

Bisected (release CLI per commit): `can_shell` → `31302718b` (#2769), `wall_hook` → `2d382185b` (#2863), missing spheres → `608cb820a` (#2708).

3 Test runs

Command	Result	Duration
`cargo build --release -p cad-cli --no-default-features` before / after	**ok, 0 warnings**	5:48 / 5:19
`cargo test --workspace --exclude cad-app --exclude cad-cli`	646 passed, 1 failed (colour volumes compared bit-exact; fixed, rerun 7/7 **ok**)	14:44
`cargo test -p cad-cli --no-default-features`	**110 passed, incl. `solidcore_against_the_reference`**	23:03
`cargo test -p cad-cli usage`	**7 passed**	0.5 s
clippy `-D warnings` (workspace headless, cad-cli), `cargo fmt --check`	**clean**	2:07
`cargo +1.98.0 check -p cad-wasm --target wasm32-unknown-unknown`	**ok**	2:00
`tests/everyday/check.sh --bugs` before / after	19/19 → **18/19** (wall_hook); known bugs fixed 10 → 11, open 7 → 6	1:38 / 2:01

Not run: `cad-app` (no WebKitGTK), web unit and e2e tests, `compat.sh` (no OpenSCAD), `tests/step/verify.py` (no OpenCASCADE), Wasm build and sizes.

4 Comparison with SMLib's frozen results

`mcad diff-reference tests/kernel-reference.json --mesh`, release, 101 files, 2604 nodes.

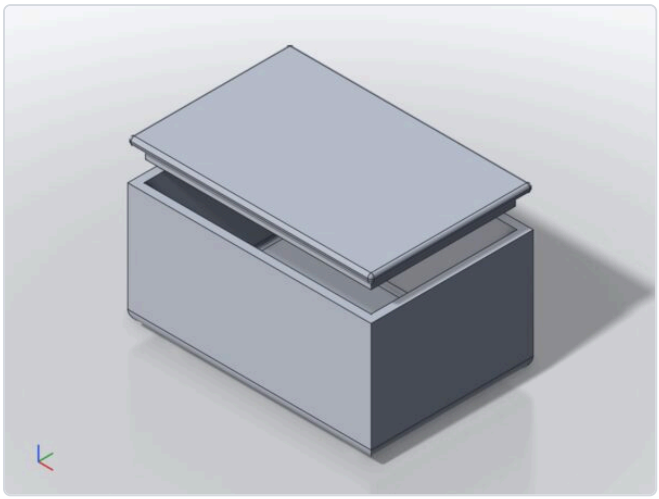
Run	same	unsupported	blocked	differ	failed	eval s
before (`f4f05b1`)	2596	3	3	2	0	120.4
new pin, only the polyhedron port	2587	3	8	4	2	138.8
final (+ shell retry, + ball patch → sphere)	**2590**	3	6	4	1	139.1

Model / node	Before → final	Cause
`wall_hook`: 14 union; 13 difference, 42 wall_hook, 1 group	same → **failed**; same → blocked	regression #2863
`f12_shell_loft_square_circle`: 6 shell, 1 group	same → **differ**	gain: SMLib's shell fails, solidcore builds 58 217 mm³, walls 2.00 mm
`can_shell`: 9 shell, 20 can, 1 group	same → failed (new pin) → **same**	#2769, fixed by the shell retry

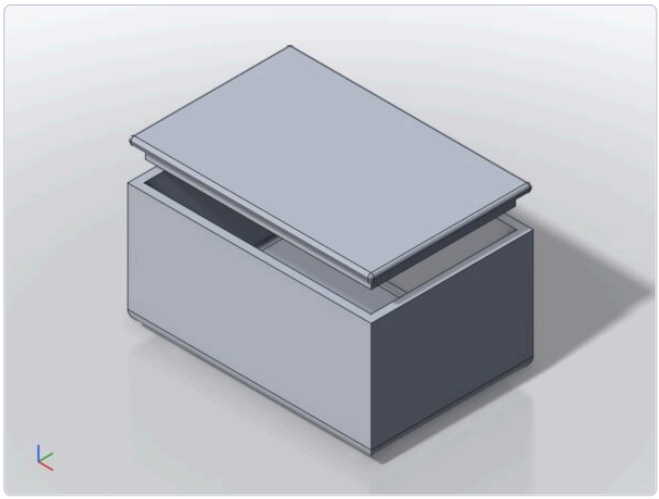
Meshes at 0.05 mm: 100 ok, 100 watertight, chord max 0.0500 mm in both runs. 33 of 101 files mesh differently, 13 with other triangle counts.

5 Renders

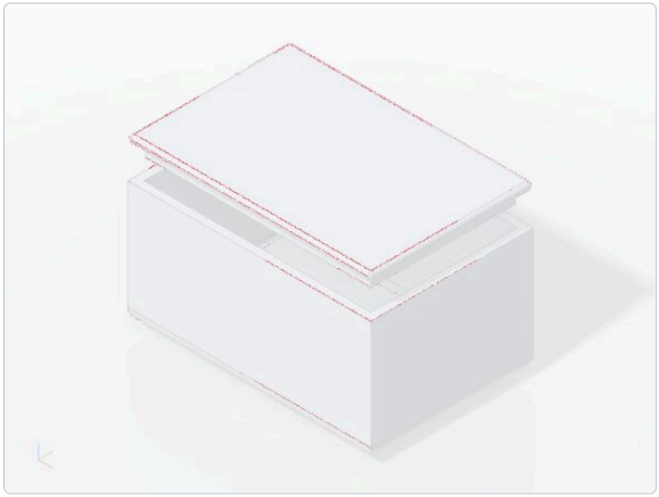
Studio renders, 900 × 675, before and after. Pixels differing by more than 15/255: bracket, macropad, flashlight, gear 0; tray 22; mug 91; box-lid 1087 (0.18 %, only anti-aliasing on edges).



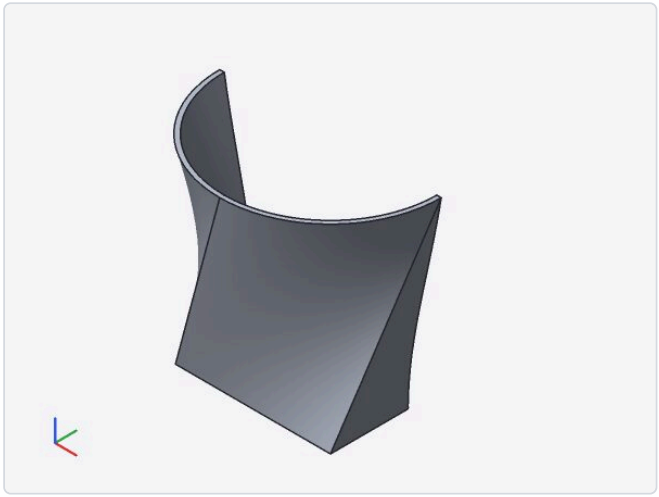
05-box-lid, before



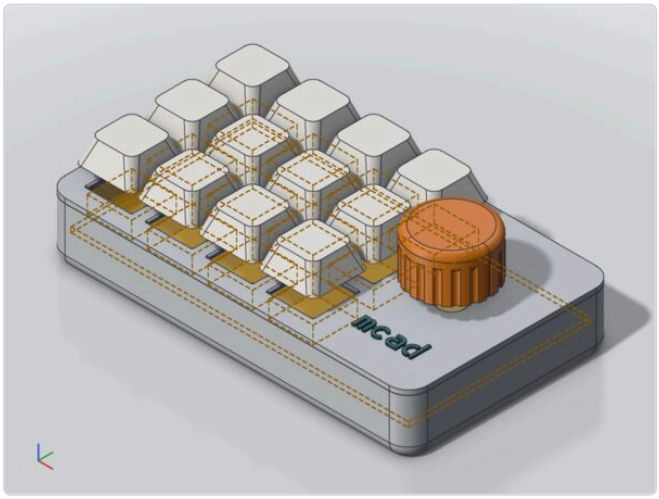
05-box-lid, after



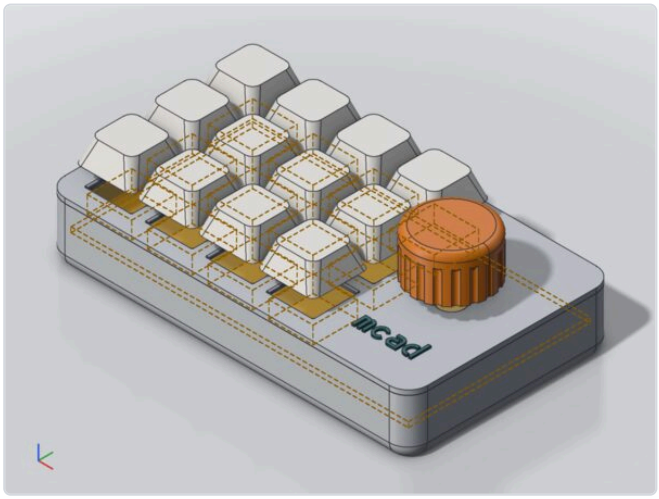
Difference box-lid (red: pixels that differ), only edge pixels



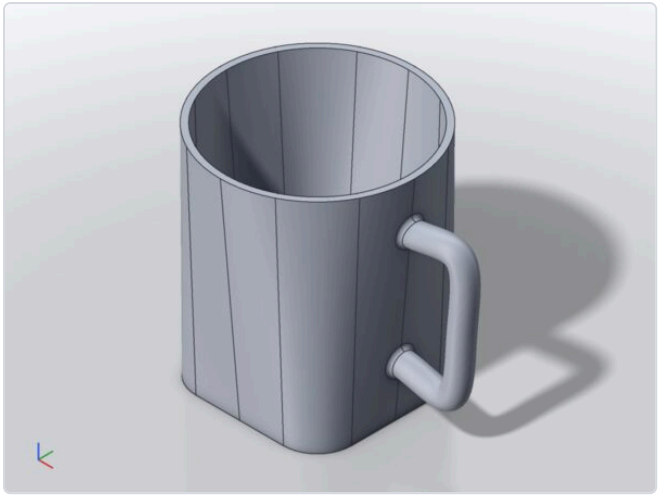
f12_shell_loft_square_circle after: shell builds (section, wall 2.00 mm)



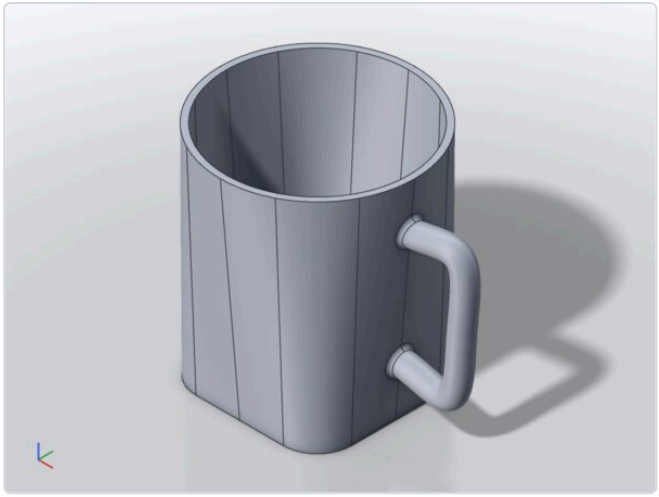
08-macropad, before



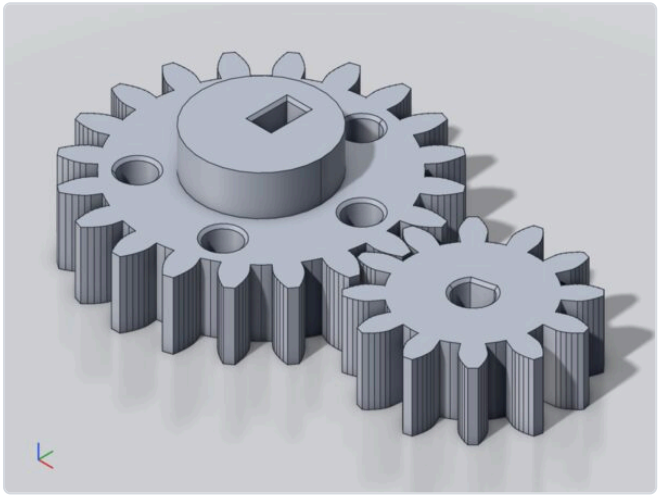
08-macropad, after (identical)



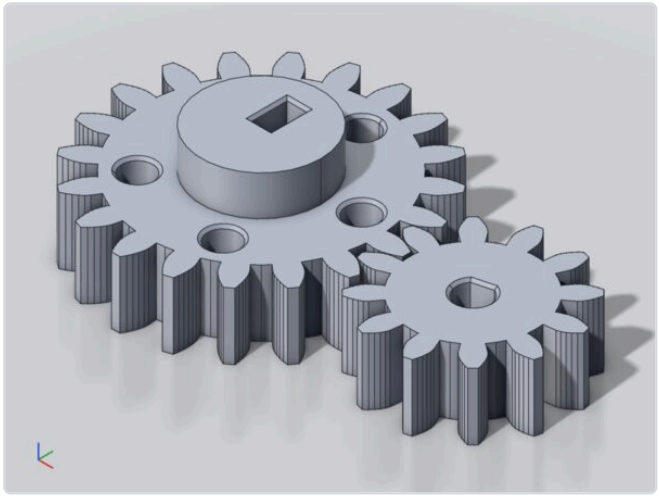
07-mug, before



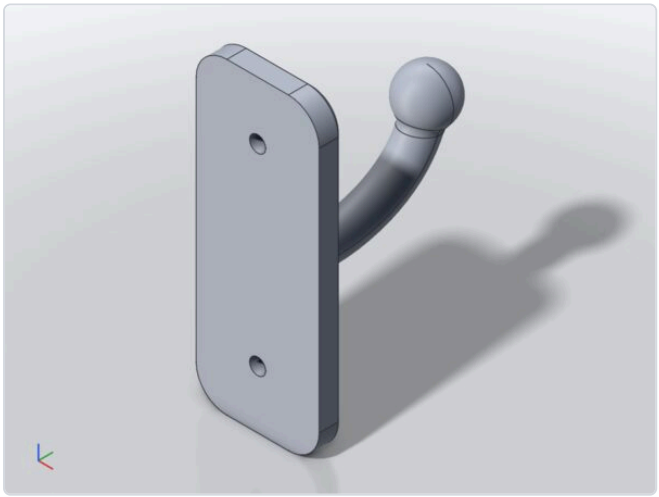
07-mug, after (91 pixels differ)



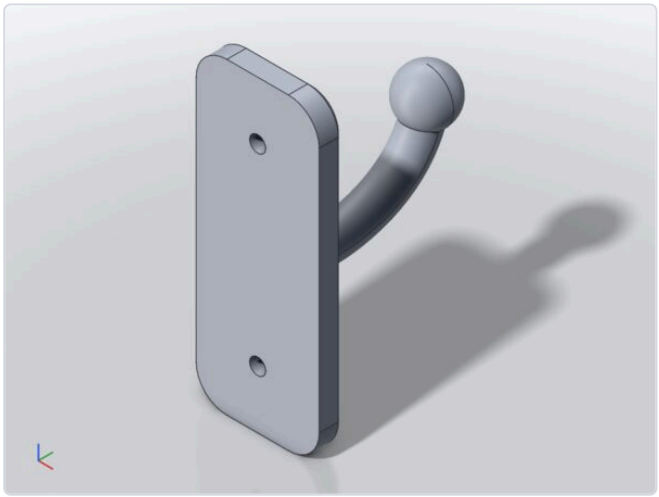
everyday/gear, before



everyday/gear, after (identical)



everyday/wall_hook, before: fillet 2.5 between tube and knob



everyday/wall_hook, after: fillet fails, union without rounding

```
tests/models/everyday/wall_hook.mcad:14:9: error (kernel): fillet failed (2.5 mm): solidcore: fillet:
solidcore returned a body that is not closed (faces missing): the blends of neighbouring edges collide,
size too large for the wall between them?
```

6 Timings

`mcad eval --json`, release, no daemon, median of 3 runs, old and new binary alternating. Models under 0.5 s vary up to 1.6× between runs.

Model	Nodes	Before s	After s	Ratio
01-plate	12	0.28	0.38	1.36
02-bracket	7	0.35	0.44	1.27
03-tray	11	0.80	1.19	1.49
04-hook	5	0.23	0.29	1.26
05-box-lid	29	0.81	0.70	0.87
06-name-tag	15	0.64	0.60	0.94
07-mug	11	7.93	8.61	1.09
08-macropad	253	14.26	15.13	1.06
09-flashlight	35	0.86	1.03	1.20
spec/text_sign	51	14.72	11.51	0.78
everyday/key_rack	68	13.02	13.26	1.02
everyday/trivet	63	9.44	16.12	1.71
everyday/mug	13	9.22	6.57	0.71
everyday/gear	48	4.93	5.53	1.12
everyday/housing	126	3.95	4.88	1.23
project-housing	33	0.38	0.40	1.06
spec/can_shell	9	0.99	2.19	2.22
everyday/wall_hook (fails after)	31	3.11	2.69	0.87
Total		**85.92**	**91.52**	**1.065**

`can_shell` is slower because the shell now runs twice (first attempt fails). `wall_hook` is only faster because it stops at the failure.

7 Open points

#	Point	Proposal
1	**wall_hook fillet regression (upstream #2863, `2d382185b`); 2.25 mm builds, 2.5–3 mm fail**	report upstream with the reproducer (sweep along a spline + sphere, `union(fillet = 2.5)`); keep the exception until fixed
2	Loosened check: shell volumes of rounded boxes 2.5e-6 … 3.3e-6 off the exact value (1e-8 before), tolerance in `kernel_fixes.rs` 1e-6 → 5e-6; cause upstream not isolated	review; revert the hunk if a red test is preferred until explained
3	Shell retry is a workaround for #2769. Side effect: `shell(1) cube(10, $fillet = 2)` now builds (414.96 vs SMLib 414.97), but its STL export at 0.01 mm fails in `cad-tess` (one triangle against the surface)	separate fix in `cad-tess`
4	Evaluation +6.5 % (18 models), +16 % (corpus); trivet ×1.71	profile trivet against the new pin
5	f12 source comment still says "open"; gain not bisected (between `e3a41166d` and `dbab64b81`)	update together with the next reference freeze

Data: `reports/tmp/report24/` (not checked in): binaries before/after, `diff-reference.json` per run, timings, renders, bisect logs, reproducers.